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Supplementary Materials for

- **Intermittent non-canonical nitrification is carbon-regulated in emerging pathogens.**

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This PDF file includes:

Supplementary Text
Figs. S1, S2, and S3
Tables S1, S3, S4, S11, S15, S17, S18, S19, S20
Legends for Tables S2, S6, S7, S8, S9, S10, S12, S13, S14, S16

Supplementary Text

Results

Approach and expansion of the genomics repertoire of *Achromobacter xylosoxidans*

Several studies have confirmed the ability of *Ax* to perform heterotrophic nitrification (*1*, *2*). This process involves simultaneously generating energy from an organic carbon source while oxidizing ammonium. However, little is known about the enzymes involved in this process for heterotrophic nitrifiers. Most of the 133 genomes of *Ax* strains contained the genes that encode for the enzymes that catalyze the oxidation of NH_4^+ to hydroxylamine (Ammonia monooxygenase, Amo) and the oxidation of nitrite to nitrate (Nitrate reductase, Nap) (**Fig. 2F**). Nonetheless, there is limited information on the enzymes responsible for oxidizing hydroxylamine (NH_2OH) to nitrite. A previous study proposed a nitrification pathway for the heterotrophic nitrifier *Alcaligenes mmonioxydans* sp (*3*). The genera Alcaligenes and Achromobacter are closely related, with multiple species historically reassigned between the two (*4*, *5*). Therefore, *Alcaligenes ammonioxydans* sp. can potentially share metabolic capabilities with *A. xylosoxidans*. Wu MR and colleagues (*3*) suggested that *A. ammonioxydans* condensates hydroxylamine (ham) to pyruvic oxime spontaneously, followed by the oxidation of ham to nitrite via pyruvic oxime dioxygenase (Pod) (**Fig. 4A**). Therefore, through BLASTp, we first identified in *A. xylosoxidans* G1401 (strain used for iGC1310 reconstruction) a homolog of the Pod enzyme from *A. ammonioxydans* (**Table S5**). This sequence was then used as a query to detect Pod across the *Ax* pangenome. **Fig. 2F** shows that 132 of 133 Ax strains encode Pod. The presence of all proteins involved in heterotrophic nitrification (Amo, Pod, and Nap) in nearly all strains suggests that this pathway has been evolutionarily conserved.

Enhancing the predictability capabilities of the GEM using experimentally determined data of the biomass objective function and high-throughput phenotyping

The Biomass Objective Function (BOF) is a stoichiometric reaction that represents *Ax* biomass composition, including proteins, carbohydrates, lipids, nucleic acids, and other macromolecules (*6*). The stoichiometric value of each metabolite is defined as millimoles per gram dry weight (mmol/gDW) of biomass.

Initially, the BOF was imported from the *i*ML1515 template model and was further adapted to *Ax* traits. The total mass fractions of DNA, carbohydrates, lipids, and proteins in *Ax* biomass were calculated experimentally, accounting for 75.31% of the total dry cell weight (3.8%, 4.9%, 13.27%, and 53.34%, respectively) (**Fig. 3A**). These values are consistent with previous reports for *E. coli*, supporting *i*ML1515 as a suitable template (*7*).

Remaining macromolecules, such as RNA, cofactors, murein, lipopolysaccharides (LPS), and inorganic ions, were normalized to the residual 24.69% of the biomass, while maintaining the relative proportions among them as defined in the *E. coli* model (*i*ML1515) (**Fig. 3A**). Amino acid and fatty acid profiles were determined experimentally. In contrast, nucleic acid profiles were derived from the *Ax* genome. Growth-Associated Maintenance energy (GAM, 75.38

mmol/gDW ATP) and Non-Growth-Associated Maintenance energy (NGAM, 6.86 mmol/gDW ATP) were imported from *i*ML1515. The updated BOF was set as the objective function for all the simulations mentioned in this study.

To our knowledge, this BOF remains the only characterization of *Ax* biomass composition available.

Genomic and metabolic modeling analyses reveal a persistent nitrification pathway in *Achromobacter xylosoxidans* regulated by carbon availability

In order to understand the conditions that involve the activation/deactivation of heterotrophic nitrification in *Ax*, we manually added to the *i*GC13101 model all the biochemical reactions associated with the nitrifying enzymes. The proposed pathway of heterotrophic nitrification in *A. xylosoxidans* is described in **Fig. 4A**.

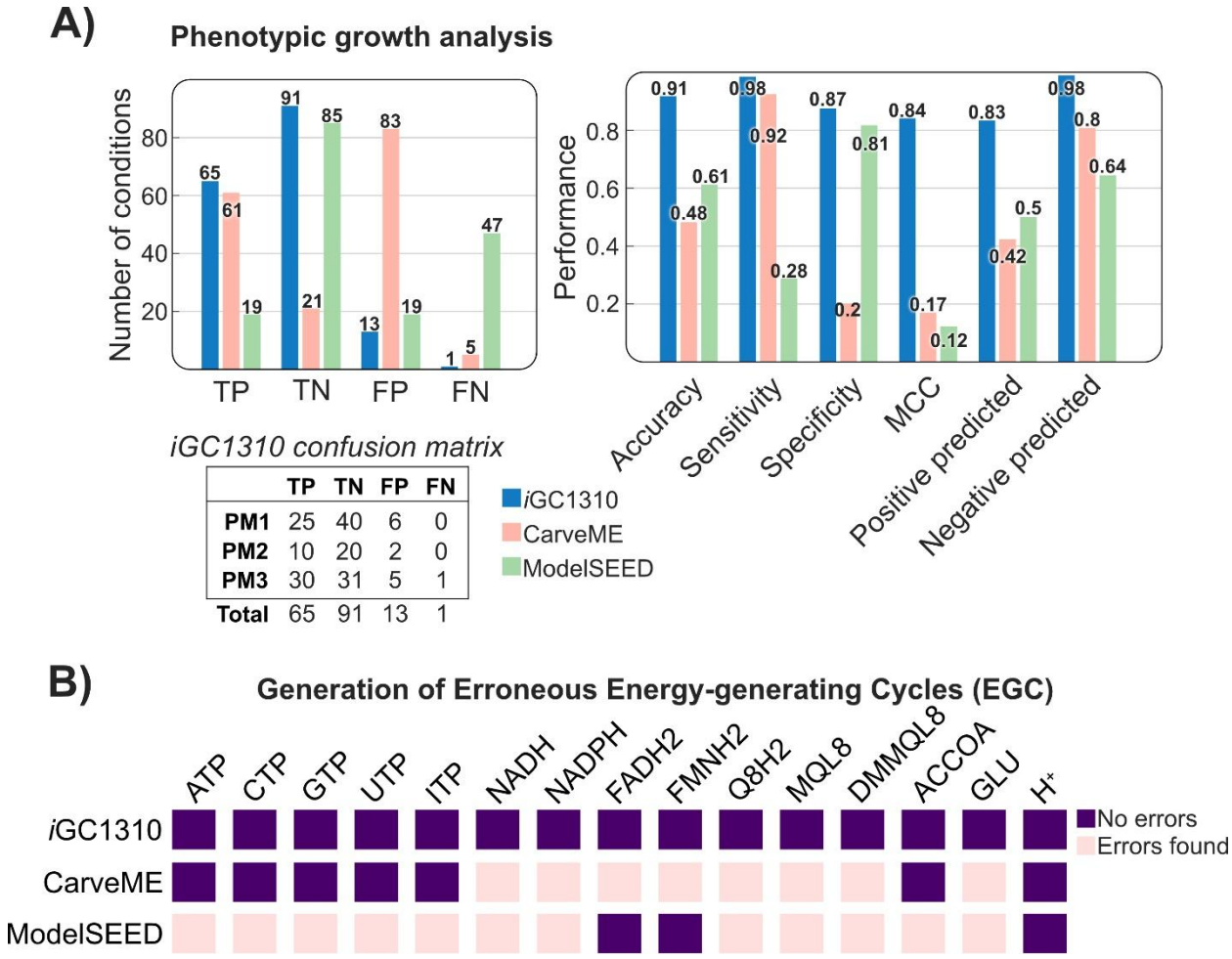
To elucidate how *Ax* activates the nitrification pathway, we performed robustness analysis simulations using low and high succinate uptake rates (0-4 mmol/g DW/h), which can be correlated with low and high concentrations (**Fig. 4A**). The *i*GC1310 model was constrained in minimal medium with NH_4^+ uptake rate fixed at 1 mmol/g DW/h. **Fig. 4A** shows that at low succinate uptake rates (~0.5 - ~2.5 mmol/gDW/h), the nitrifying enzymes are active. Ammonium is consumed and oxidized to hydroxylamine (NH_2OH) by the catalysis of Amo, followed by a spontaneous condensation reaction between hydroxylamine and pyruvate, which produces pyruvic oxime. Subsequently, pyruvic oxime is oxidized to nitrite through Pod, which is further oxidized to nitrate by Nap. The ubiquinol (QH_2) generated during the nitrification is used by the cytochrome bc1 complex (complex III) to reduce a molecule of cytochrome c that is further utilized by the terminal oxidase (cytochrome c oxidase, COX) to reduce oxygen (final electron acceptor) to water. This process generates a proton gradient between the cytoplasm and the periplasm, which induces the production of ATP by ATP synthase.

However, under higher succinate uptake rates (~2.5-4 mmol/g DW/h), the simulations predicted that the nitrification pathway is completely deactivated. The succinate dehydrogenase, which is coupled with the TCA cycle, produces all the ubiquinol required by complex III and COX. The activity of these last two reactions generates the proton motive force required for ATP synthesis.

Therefore, we experimentally measured nitrification activity in the *Ax* strains isolated from CF patients, G1401 (the strain used for *i*GC1310 modeling), G1324, and G1394**.** All these strains contained the enzymes involved in nitrification. We cultured the isolates in minimal media supplemented with ammonium chloride (1 g/L) and succinate (2.81 g/L). Triplicate cultures of *Ax* (200 mL each, initial OD600 of 0.1) were incubated at 37°C and shaken at 80 rpm for 181 hrs. Samples were taken every 18 h. and OD600, as well as ammonium nitrogen, nitrite, and protein concentrations, were determined.

As shown in **Fig. 5A and Fig. S2**, all *Ax* strains consumed NH_4^+ during the first 24 hours, which coincides with an increase in both cell density (OD_{600}) and protein concentration. After 24 hours, cell density and protein production plateau, indicating the onset of the stationary phase. Notably,

nitrite, a key metabolite in the nitrification process, begins accumulating after 24 hours. These organisms were able to initiate nitrite production (and therefore the nitrification process) just after the exponential phase. Since the stationary phase is usually related to the lack of a carbon source (*8*), we suggest that nitrification is active when the concentration of succinate is limited, which agrees with the simulation results (**Fig. 4A**). Hence, we were able to link the nitrification with the availability of carbon in the culture medium. The simulation results may indicate a shift to an alternative energy production method via nitrification. This process allows the cell to maintain redox balance and generate ATP under carbon-limited conditions.



	TP	TN	FP	FN
PM1	25	40	6	0
PM2	10	20	2	0
PM3	30	31	5	1
Total	65	91	13	1

Fig. S1. Comparative analysis of properties for the Ax models.
CarveME and ModelSEED models were reconstructed using the gap-filling and M9 medium options to ensure growth on minimal medium. A) Benchmarking of phenotypic growth simulations by all *Ax* models, showing exceptional performance of *i*GC1310 over automatically reconstructed models. B) Identification of energy-generating cycles (EGCs) for key energy metabolites (ATP, CTP, GTP, UTP, ITP, NADH, NADPH, flavin adenine dinucleotide ($FADH_2$), flavin mononucleotide (ACCOA), L-glutamate (GLU), and proton exchange between periplasm and cytosol (H^+)) in *Ax* models. Dissipation reactions for each metabolite were added to all models, and EGC formation was assessed following the protocol of Fritzemeier and collegues (2017) (30). Purple indicates the absence of EGCs for a specific metabolite, while pink indicates the presence of EGC generation.

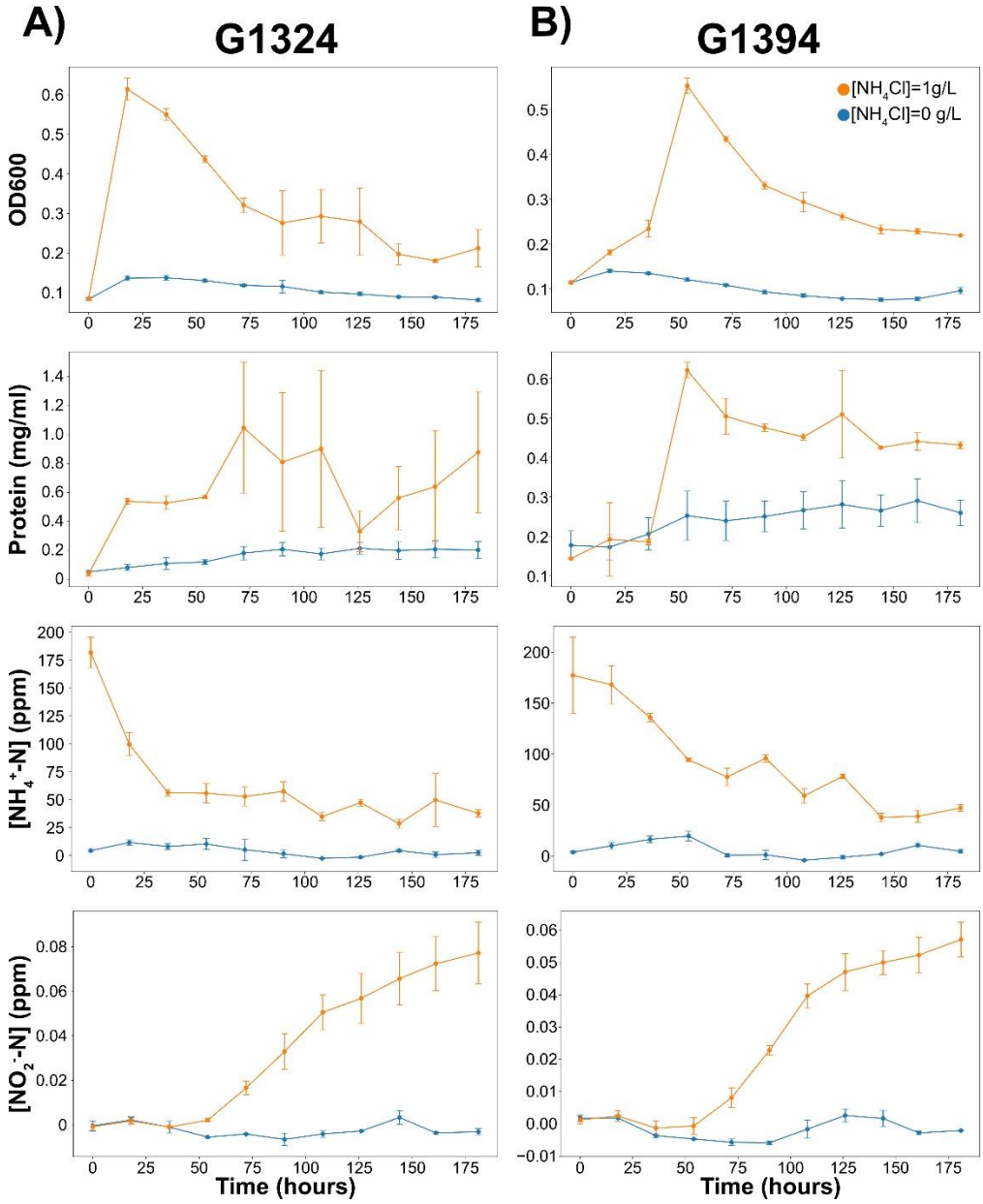


Fig. S2. Kinetic curves of *A. xylosoxidans* strains. A) Kinetic curves of G1324 strain. (B) Kinetic curves of G1394 strains. Both strains were cultured in minimal media supplemented with ammonium chloride (NH4Cl) and succinate.

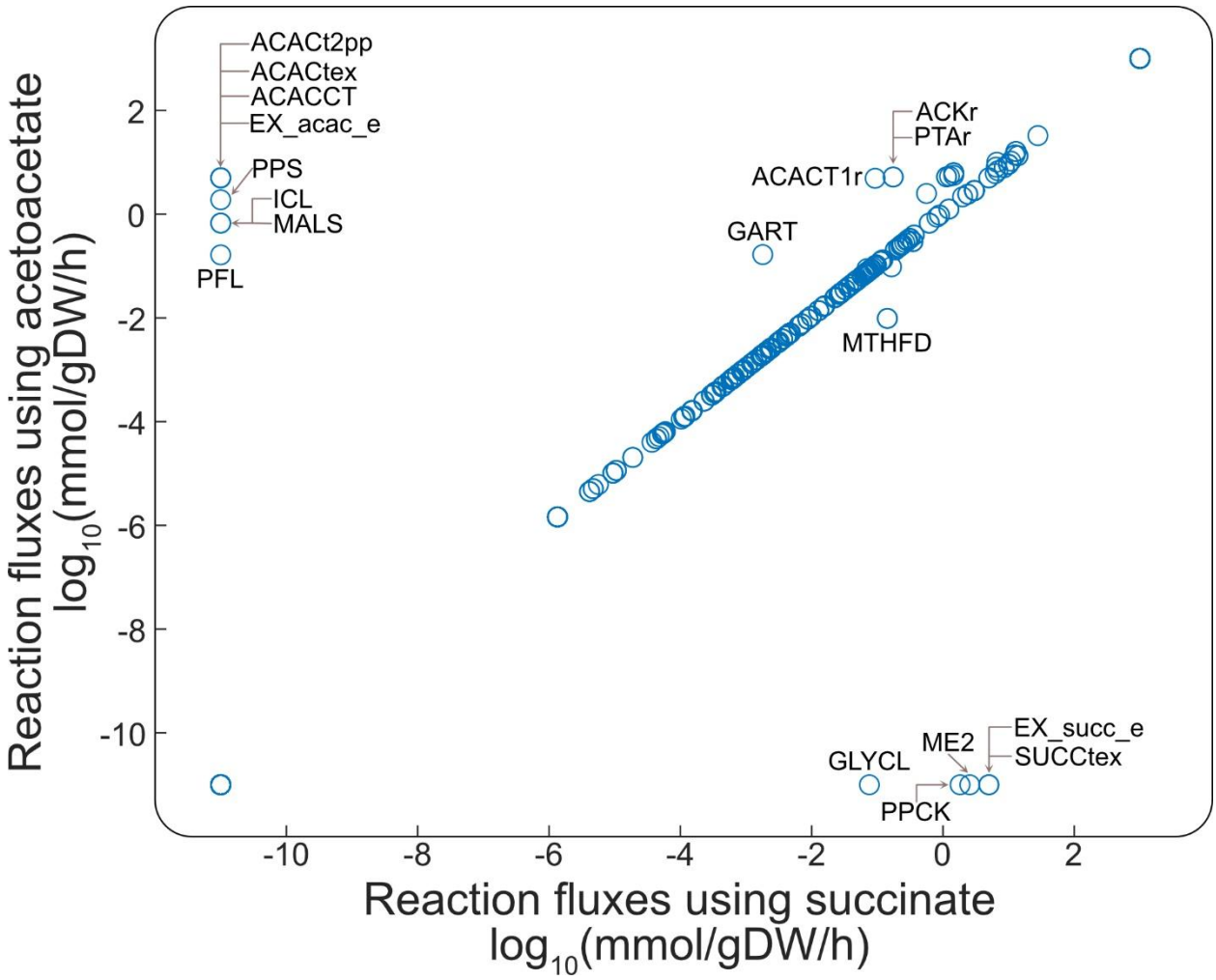


Fig. S3. Distribution of metabolic fluxes using acetoacetate or succinate as carbon sources. Ammonium was used as a nitrogen source for both simulations. Uptake fluxes for both the carbon and nitrogen sources were set at 5 mmol/gDW/h. Abbreviations: ACAC2pp, acetoacetate transport via proton symport (periplasm); ACACtex, Acetoacetate transport via diffusion (extracellular to periplasm), ACACCT, Acetyl-CoA:acetoacetyl-CoA transferase; EX_acac_e, Acetoacetate exchange; PPS, Phosphoenolpyruvate synthase; ICL, Isocitrate lyase; MALS, Malate synthase; PFL, Pyruvate formate lyase; GART, N1-(5-phospho-beta-D-ribosyl)glycinamide ligase (ADP-forming); ACACT1r, Acetyl-CoA C-acetyltransferase; PTAr, Phosphotransacetylase; ACKr; Acetate kinase; MTHFD, Methylenetetrahydrofolate dehydrogenase (NADP); GLYCL, Glycine Cleavage System; ME2, Malic enzyme (NADP); PPCK, Phosphoenolpyruvate carboxykinase; EX_succ_e, Succinate exchange; SUCCtex, Succinate transport via diffusion (extracellular to periplasm)

Table S1. Diseases associated with *Achromobacter xylosoxidans* in humans

Disease	Context	Reference
Cystic Fibrosis respiratory infections	Chronic colonization and persistent infection in the lungs of CF patients. Associated with lung function decline.	(*9–11*)
Pneumonia	Coarse bilateral crackles. Chronic infection.	(*12, 13*)
Meningitis	Rare CNS infection, often post-trauma or catheter-related; high morbidity and mortality	(*14–16*)
Endocarditis	Associated with heart native and prosthetic valves or devices. High crude mortality (>50%)	(*17, 18*)
Urinary Tract Infections	Very rare infection, usually in patients with urological abnormalities, malignancy, diabetes, renal failure, or transplant	(*19–21*)
Peritonitis	Notably in patients on peritoneal dialysis. Often presents with abdominal pain, cloudy dialysate, or septic shock. High risk of relapse and treatment failure due to multidrug resistance and biofilm formation on catheters	(*22–24*)
Osteomyelitis	Post-traumatic or post-surgical cases are described. Cases reported in diabetic and immunocompetent patients. Often chronic. Treatment usually involves surgical debridement or amputation plus prolonged antibiotics	(*25, 26*)
Ocular infections	Infection associated with contact lens wear, prior corneal surgery (LASIK, keratoplasty), or steroid use. Usually indolent, recurrent, and drug-resistant. Severe cases may require keratoplasty	(*27–29*)
Otitis	*Ax* was first identified in ear discharge from patients with chronic otitis media. Often, multidrug-resistant strains. Occurs in both immunocompromised and immunocompetent patients	(*30–32*)

Table S3. Details of isolated *Ax* strains obtained from patients with cystic fibrosis.

Strain #	Patient	Patient location	Sequencing method
G1318	DC32	San Diego CA	Short read / Illumina
G1323	DC32	San Diego CA	Short read / Illumina
G1324	DC32	San Diego CA	Long read/ ONT*
G1325	DC32	San Diego CA	Short read / Illumina
G1387	PCF1	Rhode Island	Long read/ ONT
G1388	PCF1	Rhode Island	Short read / Illumina
G1394	DC32	San Diego CA	Hybrid
G1401	PCF1	Rhode Island	Long read/ ONT
G1407	EH1	Atlanta GA	Short read / Illumina
G1408	EH1	Atlanta GA	Short read / Illumina
G1409	EH1	Atlanta GA	Short read / Illumina
G1410	EH1	Atlanta GA	Short read / Illumina
G1412	SCaPH	South Carolina	Long read/ ONT
G1413	SCaPH	South Carolina	Long read/ ONT
G1414	SCaPH	South Carolina	Long read/ ONT
G1417	PCF1	Rhode Island	Hybrid
G1418	PCF1	Rhode Island	Hybrid
G1423	PCF1	Rhode Island	Long read/ ONT

This project strain designation
* Oxford Nanopore Technologies

Table S4. Results of BUSCO analysis in genomes of isolates obtained from clinical isolates of patients with cystic fibrosis

Isolate	Completeness	Single	Duplicated	Fragmented	Missing	GenomeSize(pb)
G1410	99.2	98.5	0.7	0.4	0.4	6324016
G1432	77.7	77.6	0.1	15	7.3	6749065
G1413	74.4	73.8	0.6	16	9.6	6206895
G1388	98.6	98	0.6	0.4	1	6284599
G1408	99.2	98.5	0.7	0.4	0.4	6321513
G1407	99.2	98.5	0.7	0.4	0.4	6319065
G1409	99.2	98.5	0.7	0.4	0.4	6334465
G1418	99.3	98.7	0.6	0.3	0.4	6290280
G1414	85	84	1	9.6	5.4	6207369
G1325	98.5	98.1	0.4	0.6	0.9	6449526
G1412	76.3	76	0.3	13.5	10.2	6206928
G1401	93.8	93.2	0.6	3.9	2.3	6293570
G1324	66.8	66.7	0.1	20.9	12.3	6414488
G1323	98.7	97.8	0.9	0	1.3	6625099
G1387	98.6	98	0.6	0.6	0.8	6227005
G1433	78	77.9	0.1	14.8	7.2	6749792
G1434	74.5	74.4	0.1	17.6	7.9	6748899
G1394	98.7	98.4	0.3	0.6	0.7	6513437
G1417	99	98.4	0.6	0.4	0.6	6289541
G1423	79.2	78.6	0.6	13.5	7.3	6277991
G1318	98.7	98.3	0.4	0.7	0.6	6465045

Table S5. Nitrifying reactions in *i*GC1310 with their associated genes.

Rection ID in model	Reaction name	Reaction abbreviation in Figure 4	Reaction formula	Gene
AMO	Ammonia monooxygenase	AMO	2 h_c + nh4_p + q8h2_c + o2_p -> h2o_p + 3 h_p + q8_c + ham_p	222.273.peg.6062
PYROXIMA	Pyruvate oximation	PYROXIMA	pyr_p + h_p + ham_p -> h2o_p + pyrox_p	Spontaneous
POD	Pyruvic oxime dioxygenase	POD	o2_p + pyrox_p -> pyr_p + 2 h_p + no2_p	222.273.peg.2420
NO3R1bpp	Respiratory nitrate reductase	NAP_ABDE	h2o_p + no2_p + q8_c <=> q8h2_c + no3_p	222.273.peg.2781 and 222.273.peg.2782 and 222.273.peg.2783 and 222.273.peg.2784
NADH16pp	NADH dehydrogenase	nadh16pp	4 h_c + q8_c + nadh_c -> 3 h_p + nad_c + q8h2_c	222.273.peg.1441
SUCDi	Succinate dehydrogenase	SUCDi	q8_c + succ_c -> q8h2_c + fum_c	222.273.peg.1484 and 222.273.peg.1485 and 222.273.peg.1482 and 222.273.peg.1483 and 222.273.peg.1486 and 222.273.peg.5623
bc1com	Cytochrome bc1 complex III	bc1com	2 h_c + q8h2_c + 2 cytcoxi_p -> 4 h_p + q8_c + 2 cytcred_p	222.273.peg.202 and 222.273.peg.203 and 222.273.peg.204
CytCOX	Cytochrome c oxidase terminal oxidase	COX	4 h_c + 0.5 o2_c + 2 cytcred_p -> 2 h_p + h2o_c + 2 cytcoxi_p	222.273.peg.255 and 222.273.peg.256 and 222.273.peg.259 or ((222.273.peg.4597 or 222.273.peg.4416) and (222.273.peg.4417 or 222.273.peg.4596) and (222.273.peg.4418 or 222.273.peg.4595) and (222.273.peg.4419 or 222.273.peg.4594))

Table S11. Summary of studies linking nitrification and denitrification activity to succinate as a carbon source

Organism/system	Biochemical process	Succinate role	Key finding	Reference
Achromobacter spiritinus HS2	Nitrification and denitrification	Sodium succinate was the optimal carbon source for growth and nitrogen removal	With succinate, *A. spiritinus*HS2 achieved up to 99.1% ammonium removal and 95.5% total nitrogen removal at 8 °C via heterotrophic nitrification-aerobic denitrification, without nitrite accumulation or N_2O emissions.	(*33*)
Glutamicibacter arilaitensis EM-H8	Denitrification	Tested as a sole carbon source. Identified as optimal for nitrate (NO_3^--N) removal, reaching 95.8–97.6% efficiency	Succinate enabled efficient aerobic denitrification	(*34*)
Achromobacter xylosoxidans CF-S36	Nitrification	Succinate was the most preferred carbon source for nitrification (highest NH_4^+ removal rate, 6.3 mg/L/h). It also supported efficient nitrate and nitrite denitrification	*A. xylosoxidans* CF-S36 achieved ~98% NH_4^+ removal within 36 h using succinate; ~100% NO_3^- removal within 36 h , and ~76% NO_2^- removal within 36 h	(*1*)
Pseudomonas putida NP5	Nitrification and denitrification	Succinate was identified as the optimal carbon source for heterotrophic nitrification-aerobic denitrification and simultaneous nitrogen-phosphorus removal	With succinate, *P. putida* NP5 achieved ~96% NH_4^+ removal and ~90% PO_4^{3-} removal within 24 h; nearly complete NO_2^- (99.4%) and NO_3^- (99.6%) removal within 24 h during aerobic denitrification. Negligible NO_2^- or N_2O accumulation occurred.	(*35*)
Paracoccus denitrificans (NCIB 8944)	Denitrification	Succinate was used as the carbon source during anaerobic growth with nitrate as electron acceptor. It fueled the respiratory chain linked to denitrification.	Cells reduced nitrate → nitrite → nitrous oxide → N_2 without accumulation of intermediates. Electron flow from succinate supported complete denitrification	(*36*)
Paracoccus denitrificans PD1222	Denitrification	Succinate was supplied as the sole carbon source (30 mM) during anaerobic denitrification assays.	With succinate, *Paracoccus denitrificans* PD1222 carried out complete denitrification ($NO_3^- \rightarrow NO_2^- \rightarrow NO \rightarrow N_2O \rightarrow N_2$) without accumulation of gaseous intermediates.	(*37*)
Mixed microbial community (biofilms in PBS-packed reactors). Microbial analysis showed dominance of Azoarcus, Simplicispira,	Denitrification	Poly(butylene succinate) (PBS) acted as both solid carbon source and biofilm carrier for denitrifiers.	Stable denitrification achieved with real RAS (recirculating aquaculture system) wastewater: denitrification rates up to	(*38*)

Acidovorax, Comamonas, Paracoccus, Labrenzia, all linked to heterotrophic denitrification.		Succinate was released gradually during polymer degradation and used as electron donor for denitrification	0.53-0.66 kg NO_3^--N m^{-3} d^{-1}, with nitrite kept below 1 mg/L.	
Mixed microbial biofilm (groundwater bioreactor). The major phyla were Proteobacteria, specifically Betaproteobacteria, Burkholderiales, and Comamonadaceae the major phyla	Denitrification	Poly(butylene succinate) (PBS) acted as solid carbon source and biofilm carrier. Succinate fueled denitrifiers	Continuous operation for ~2 years achieved 88–97% nitrate removal, effluent nitrate 3.3–8.8 mg/L. Denitrification rates: 0.25–0.35 g N L^{-1} d^{-1} (20–29 °C), dropping to 0.12 g N L^{-1} d^{-1} (10–18 °C).	(*39*)

Table S15. Presence or absence of denitrification enzymes in *Ax* strains obtained from clinical isolates of patients with cystic fibrosis

Strain	Nitrate reductase	Nitrite reductase	Nitric oxide reductase	Nitrous oxide reductase
G1401	+	-	-	-
G1407	+	-	-	-
G1408	+	-	-	-
G1409	+	-	-	-
G1410	+	-	-	-
G1412	+	-	-	-
G1413	+	-	-	-
G1414	+	-	-	-
G1324	+	+	+	+
G1394	+	+	+	+
G1325	+	+	+	+
G1323	+	+	+	-
G1318	+	+	+	+
G1434	+	+	+	+
G1433	+	+	+	+
G1434	+	+	+	+
G1387	+	-	-	-
G1388	+	-	-	-
G1417	+	-	-	-
G1418	+	-	-	-
G1423	+	-	-	-

Table S17. Presence or absence of denitrification enzymes in *Ax* strains obtained from clinical isolates of patients with cystic fibrosis

Compound	**Formula**	**Concentration (g/L)**
Ferrous sulfate heptahydrate	$FeSO_4 \cdot 7H_2O$	9.13
Calcium chloride dehydrate	$CaCl_2 \cdot 2H_2O$	7.33
Manganese(II) chloride tetrahydrate	$MnCl_2 \cdot 4H_2O$	7.95
Zinc sulfate heptahydrate	$ZnSO_4 \cdot 7H_2O$	39.17
Copper(II) sulfate	$CuSO_4$	1.57
Cobalt(II) chloride hexahydrate	$CoCl_2 \cdot 6H_2O$	2.95
Ammonium molybdate(IV) tetrahydrate	$(NH_4)_6Mo_7O_{24} \cdot 4H_2O$	1.16

Table S18. Model constraints for carbon utilization simulations

Reaction ID	Reaction name	Reaction Formula	Metabolite	Lower bound	Upper bound
EX_pi_e	Phosphate exchange	pi_e <=>	pi_e	-1000	1000
EX_co2_e	CO2 exchange	co2_e <=>	co2_e	-1000	1000
EX_fe3_e	Fe3+ exchange	fe3_e <=>	fe3_e	-1000	1000
EX_h_e	H+ exchange	h_e <=>	h_e	-1000	1000
EX_mn2_e	Mn2+ exchange	mn2_e <=>	mn2_e	-1000	1000
EX_succ_e	Succinate exchange	succ_e <=>	succ_e	0	1000
EX_fe2_e	Fe2+ exchange	fe2_e <=>	fe2_e	-1000	1000
EX_zn2_e	Zinc exchange	zn2_e <=>	zn2_e	-1000	1000
EX_mg2_e	Mg exchange	mg2_e <=>	mg2_e	-1000	1000
EX_ca2_e	Calcium exchange	ca2_e <=>	ca2_e	-1000	1000
EX_ni2_e	Ni2+ exchange	ni2_e <=>	ni2_e	-1000	1000
EX_cu2_e	Cu2+ exchange	cu2_e <=>	cu2_e	-1000	1000
EX_sel_e	Selenate exchange	sel_e <=>	sel_e	-1000	1000
EX_cobalt2_e	Co2+ exchange	cobalt2_e <=>	cobalt2_e	-1000	1000
EX_h2o_e	H2O exchange	h2o_e <=>	h2o_e	-1000	1000
EX_mobd_e	Molybdate exchange	mobd_e <=>	mobd_e	-1000	1000
EX_so4_e	Sulfate exchange	so4_e <=>	so4_e	-1000	1000
EX_nh4_e	Ammonia exchange	nh4_e <=>	nh4_e	-5	1000
EX_k_e	K+ exchange	k_e <=>	k_e	-1000	1000
EX_na1_e	Sodium exchange	na1_e <=>	na1_e	-1000	1000
EX_cl_e	Chloride exchange	cl_e <=>	cl_e	-1000	1000
EX_o2_e	O2 exchange	o2_e <=>	o2_e	-1000	1000
EX_tungs_e	Tungstate exchange	tungs_e <=>	tungs_e	-1000	1000
EX_slnt_e	Selenite exchange	slnt_e <=>	slnt_e	-1000	1000

Table S19. Model constraints for nitrogen utilization simulations

Reaction ID	Reaction name	Reaction Formula	Metabolite	Lower bound	Upper bound
EX_pi_e	Phosphate exchange	pi_e <=>	pi_e	-1000	1000
EX_co2_e	CO2 exchange	co2_e <=>	co2_e	-1000	1000
EX_fe3_e	Fe3+ exchange	fe3_e <=>	fe3_e	-1000	1000
EX_h_e	H+ exchange	h_e <=>	h_e	-1000	1000
EX_mn2_e	Mn2+ exchange	mn2_e <=>	mn2_e	-1000	1000
EX_succ_e	Succinate exchange	succ_e <=>	succ_e	-5	1000
EX_fe2_e	Fe2+ exchange	fe2_e <=>	fe2_e	-1000	1000
EX_zn2_e	Zinc exchange	zn2_e <=>	zn2_e	-1000	1000
EX_mg2_e	Mg exchange	mg2_e <=>	mg2_e	-1000	1000
EX_ca2_e	Calcium exchange	ca2_e <=>	ca2_e	-1000	1000
EX_ni2_e	Ni2+ exchange	ni2_e <=>	ni2_e	-1000	1000
EX_cu2_e	Cu2+ exchange	cu2_e <=>	cu2_e	-1000	1000
EX_sel_e	Selenate exchange	sel_e <=>	sel_e	-1000	1000
EX_cobalt2_e	Co2+ exchange	cobalt2_e <=>	cobalt2_e	-1000	1000
EX_h2o_e	H2O exchange	h2o_e <=>	h2o_e	-1000	1000
EX_mobd_e	Molybdate exchange	mobd_e <=>	mobd_e	-1000	1000
EX_so4_e	Sulfate exchange	so4_e <=>	so4_e	-1000	1000
EX_nh4_e	Ammonia exchange	nh4_e <=>	nh4_e	0	1000
EX_k_e	K+ exchange	k_e <=>	k_e	-1000	1000
EX_na1_e	Sodium exchange	na1_e <=>	na1_e	-1000	1000
EX_cl_e	Chloride exchange	cl_e <=>	cl_e	-1000	1000
EX_o2_e	O2 exchange	o2_e <=>	o2_e	-1000	1000
EX_tungs_e	Tungstate exchange	tungs_e <=>	tungs_e	-1000	1000
EX_slnt_e	Selenite exchange	slnt_e <=>	slnt_e	-1000	1000

Table S20. Model constraints for simulations of nitrification analysis

Reaction ID	Reaction name	Reaction Formula	Metabolite	Lower bound (mmol/gDW/h)	Upper bound (mmol/gDW/h)
EX_pi_e	Phosphate exchange	pi_e <=>	pi_e	-1000	1000
EX_co2_e	CO2 exchange	co2_e <=>	co2_e	-1000	1000
EX_fe3_e	Fe3+ exchange	fe3_e <=>	fe3_e	-1000	1000
EX_h_e	H+ exchange	h_e <=>	h_e	-1000	1000
EX_mn2_e	Mn2+ exchange	mn2_e <=>	mn2_e	-1000	1000
EX_succ_e	Succinate exchange	succ_e <=>	succ_e	[0-4]	1000
EX_fe2_e	Fe2+ exchange	fe2_e <=>	fe2_e	-1000	1000
EX_zn2_e	Zinc exchange	zn2_e <=>	zn2_e	-1000	1000
EX_mg2_e	Mg exchange	mg2_e <=>	mg2_e	-1000	1000
EX_ca2_e	Calcium exchange	ca2_e <=>	ca2_e	-1000	1000
EX_ni2_e	Ni2+ exchange	ni2_e <=>	ni2_e	-1000	1000
EX_cu2_e	Cu2+ exchange	cu2_e <=>	cu2_e	-1000	1000
EX_sel_e	Selenate exchange	sel_e <=>	sel_e	-1000	1000
EX_cobalt2_e	Co2+ exchange	cobalt2_e <=>	cobalt2_e	-1000	1000
EX_h2o_e	H2O exchange	h2o_e <=>	h2o_e	-1000	1000
EX_mobd_e	Molybdate exchange	mobd_e <=>	mobd_e	-1000	1000
EX_so4_e	Sulfate exchange	so4_e <=>	so4_e	-1000	1000
EX_nh4_e	Ammonia exchange	nh4_e <=>	nh4_e	-1	1000
EX_k_e	K+ exchange	k_e <=>	k_e	-1000	1000
EX_na1_e	Sodium exchange	na1_e <=>	na1_e	-1000	1000
EX_cl_e	Chloride exchange	cl_e <=>	cl_e	-1000	1000
EX_o2_e	O2 exchange	o2_e <=>	o2_e	-1000	1000
EX_tungs_e	Tungstate exchange	tungs_e <=>	tungs_e	-1000	1000
EX_slnt_e	Selenite exchange	slnt_e <=>	slnt_e	-1000	1000

Table S2. (separate file)
List of *Ax* strains from NCBI and CF clinical isolates with isolation sources

Table S6. (separate file)
Persistent genes obtained from pan-genome analysis

Table S7. (separate file)
Shell genes identified during pan-genome analysis

Table S8. (separate file)
Cloud genes identified during pan-genome analysis

Table S9. (separate file)
Properties of *Ax* multi-strain scale metabolic models

Table S10. (separate file)
Genes and their amino acid sequences per GEM in the *Ax* multi-strain network

Table S12. (separate file)
Results of phenotypic growth experiments compared with predictions from *i*GC1310, CarveME, and ModelSEED models.

Table S13. (separate file)
Genomic location of denitrifying genes across soil- and water-derived *Ax* strains

Table S14. (separate file)
GC content of denitrifying genes across soil- and water-derived *Ax* strains

Table S16. (separate file)
Binarized growth rates of all *Ax* strains from human and non-human sources using several carbon sources

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